



Smart Kanban

An AI-powered Kanban board built with **React**, **Express**, **Tailwind CSS**, and **Google Gemini AI** that helps users organize tasks, prioritize work, and improve productivity through intelligent task analysis.



🌟 Features

📋 Task Management

- Create, edit, and delete tasks
- Drag & drop tasks across columns
- Priority labels (High, Medium, Low)
- Search tasks instantly
- Filter tasks by priority
- Local storage persistence
- Responsive user interface

🤖 AI Workspace

- 🎯 Suggest task priority
- 📋 Break complex tasks into actionable steps
- 📄 Generate board summaries
- 💡 Productivity coaching
- 💬 AI-powered Kanban chat assistant

📊 Dashboard

- Task statistics
 - Progress overview
 - Clean and modern UI
 - Responsive layout
-

🔧 Tech Stack

Frontend

- React
- Vite

- Tailwind CSS
- JavaScript (ES6+)

Backend

- Node.js
- Express.js

AI

- Google Gemini API

Deployment

- Frontend: Vercel
- Backend: Render

Project Structure

```
smart-kanban
├── public/
├── server/
│   └── server.js
├── src/
│   ├── assets/
│   ├── components/
│   ├── services/
│   ├── App.jsx
│   └── main.jsx
├── package.json
├── tailwind.config.js
└── README.md
```

AI Features

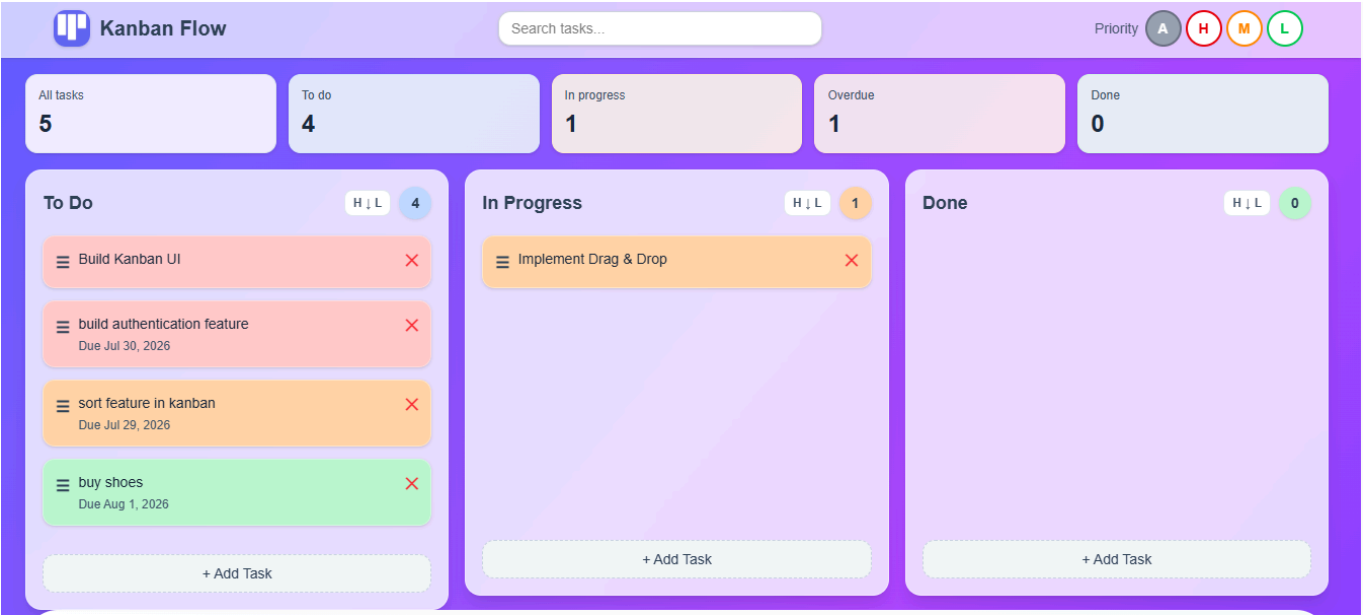
The AI assistant understands your current Kanban board and can:

- Analyze task priority
- Break tasks into implementation steps
- Summarize project progress
- Identify risks
- Suggest next actions
- Answer project-related questions
- Provide productivity guidance

The backend securely communicates with the Google Gemini API, ensuring the API key is never exposed to the client.

Screenshots

Dashboard



AI Workspace

!(/screenshots/AI_Tools.png)

AI Chat Assistant

 Kanban Flow

Priority

A

H

M

L

AI chat assistant

Ask follow-up questions in a conversational chat format.

✖ PRIORITY

build authentication feature

✖ BREAKDOWN

build authentication feature

Priority: High Reason:

This task is already designated as high priority on your board with an upcoming due date of 2026-07-30.

Task Breakdown 1. Set up API endpoints for user registration and login.

2. Design frontend login and signup form components.

3. Implement JWT or session-based state management.

Ask about your tasks...

Send

✖ COACH

build authentication feature

Observation: The task build authentication feature is currently in your Todo list with high priority and a due date of 2026-07-30. You also have Implement Drag & Drop currently in progress. Advice: To maintain a smooth workflow, avoid switching contexts. Focus on completing your current in-progress task first before taking on another high-priority item. Keeping your work-in-progress limit low will help you deliver faster with fewer errors. Next Step: Finish Implement Drag & Drop and move it to Done, then move build authentication feature into inprogress.

Ask about your tasks...

Send

Board Summary

✖ SUMMARY

build authentication feature

Board Summary Progress:

- 0 of 5 tasks completed.

- 1 task in progress and 4 tasks in To Do. Risks:

- No completed tasks yet.

- High priority items are sitting in To Do with upcoming deadlines. Next Best Action:

- Finish Implement Drag & Drop to clear current in-progress work. Regarding "build authentication feature":

- Current Status: To Do

- Priority: High

- Recommendation: Assign this as the next task after clearing in-progress work to meet the 2026-07-30 due date.

Ask about your tasks...

Send

 Installation

Clone the repository

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```
git clone https://github.com/KeerthyHQ/smart-kanban.git
cd smart-kanban
```

Install dependencies

```
npm install
```

Start the frontend

```
npm run dev
```

Start the backend

```
npm run server
```



Environment Variables

Create a `.env` file inside the project root.

```
PORT=3001
GEMINI_API_KEY=YOUR_API_KEY
GEMINI_MODEL=gemini-flash-latest
```



How It Works

```
graph TD
    A[React Frontend] --> B[Express Backend]
    B --> C[Google Gemini API]
```

The frontend sends AI requests to the Express backend.

The backend securely calls the Gemini API and returns AI-generated responses.



Future Enhancements

- User Authentication
 - Cloud Database
 - Team Collaboration
 - Due Dates & Reminders
 - Activity History
 - Dark Mode
 - Analytics Dashboard
 - Export & Import Boards
-



Key Concepts Practiced

- React Components
 - React Hooks
 - State Management
 - Drag & Drop UI
 - REST APIs
 - Express.js
 - AI Integration
 - Prompt Engineering
 - Environment Variables
 - Responsive Design
 - Local Storage
 - Component Architecture
-



Author

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Support

If you found this project useful, consider giving it a ★ on GitHub.

rt & Import Boards



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